Brine Pretreat D_i TEA

Produced-Water Lithium Extraction

TBAC/DA DES + TOPO from Smackover feed basis to staged extraction and screening TEA

Chemical engineering technical review

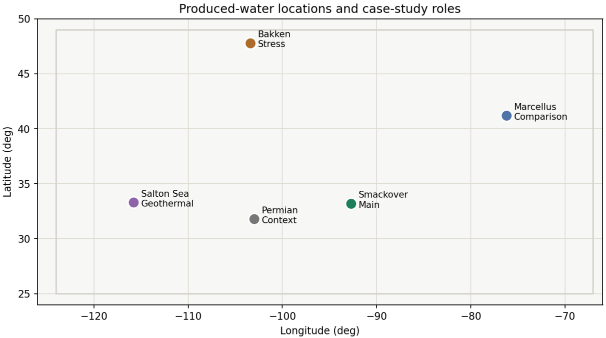
Smackover Li/Na feed basis → pretreatment → TBAC/DA DES + TOPO → ALAMO surrogate → PrOMMiS/IDAES

May 15, 2026

A pretreated Smackover Li/Na stream connects realistic produced-water chemistry to staged extraction and screening-level economics.

CASE BASIS

Smackover Is The Main Produced-Water Case



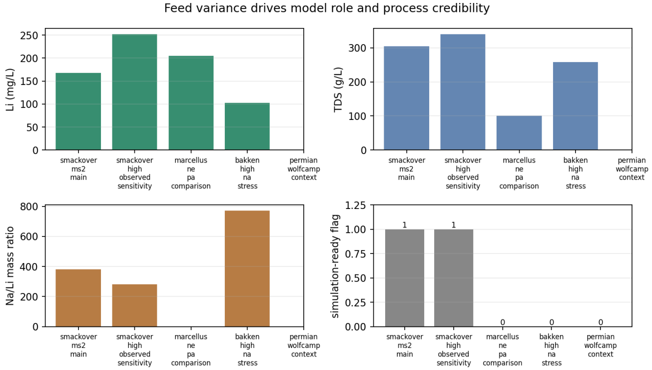
Case basis

Southern Arkansas Smackover combines meaningful lithium grade with high-salinity major-ion chemistry.

- MS-2 feed: 168 mg/L Li, 64 100 mg/L Na, and 305 000 mg/L TDS.
- High-observed Smackover sensitivity: 252 mg/L Li.
- Marcellus, Bakken, Permian, and Salton Sea remain comparison cases.

Sources: USGS Smackover data release; selected feed table.

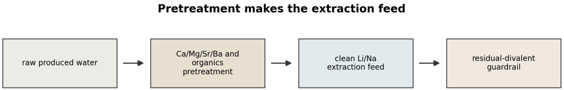
Feed Variance Sets The Model Role



- Smackover MS-2 anchors the main flowsheet and TEA case.
- The high-Li Smackover row tests feed-grade sensitivity.
- Incomplete major-ion rows stay outside the main process calculation.

Feed	Role
MS-2	main case
High-Li Smackover	sensitivity
Marcellus/Bakken	context

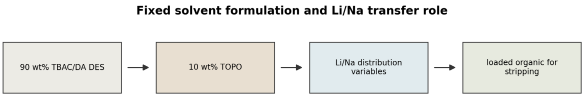
Pretreatment Creates The Li/Na Extraction Feed



- Raw produced water is not the extraction feed.
- Ca^{2+}, Mg^{2+}, Sr^{2+}, Ba^{2+}, suspended solids, oil, and organics are upstream treatment burdens.
- Lithium loss in pretreatment is carried into TEA sensitivity.

SOLVENT AND TRANSFER SURFACE

TBAC/DA DES + TOPO Is The Active Solvent



- Organic phase: 90 wt% TBAC/DA DES and 10 wt% TOPO.
- TBAC:DA molar ratio: 1:2.
- Temperature: 23 °C.
- Aqueous pH: 10.4.

Source: Rezaee DES/TOPO extraction basis and supporting information.

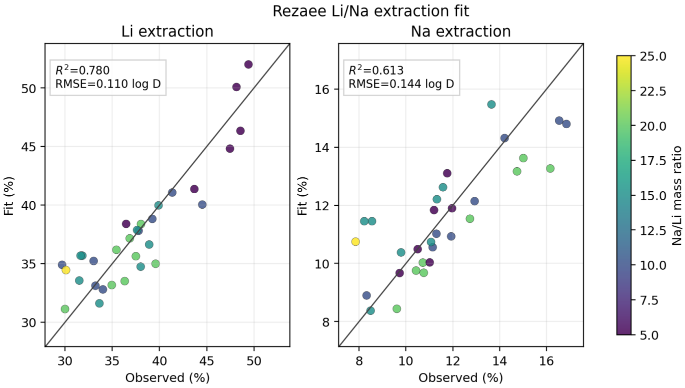
Stage-Response Surface Spans Source And Smackover Domains

Domain	Rows	Use
Rezaee DOE rows	26	Li/Na target surface
optimum rows	5	high-response anchors
real-brine row	1	high-Na/Li bound
process design	523	Smackover sensitivity

Retained source anchors	
optimized Li extraction	48.57%
optimized Li/Na selectivity	4.41
model-brine Li extraction	51.63%
model-brine Na extraction	9.97%

- The repaired package route evaluates 26 phase-tagged reactive rows.
- The surrogate uses 31 finite DOE and optimum target rows.
- Smackover MS-2 is outside the original Na/Li range and is marked as an extrapolated process case.

ALAMO-Ready Surrogate Carries The Current Li/Na Targets



- Inputs: temperature, pH, TOPO loading, and Na/Li ratio.
- Outputs: $\log D_{\mathrm{Li}}$ and $\log D_{\mathrm{Na}}$ for process transfer.
- Current target-row fit: $R^2 = 0.780$ for Li and $R^2 = 0.613$ for Na.

Output	RMSE	max error
$\log D_{\mathrm{Li}}$	0.110	0.239
$\log D_{\mathrm{Na}}$	0.144	0.367

Paper-Response Variables Feed The Process Model

$$D_i = \frac{E_i}{100 - E_i} \frac{1}{O/A}$$

$$S_{\mathrm{Li/Na}} = \frac{D_{\mathrm{Li}}}{D_{\mathrm{Na}}}$$

Rezaee extraction responses are converted to Li/Na transfer variables for process calculations

Rezaee anchors, $O/A = 1$

Optimized 10 wt% TOPO: $E_{\mathrm{Li}} = 48.57\%$, $S_{\mathrm{Li/Na}} = 4.41$, so $E_{\mathrm{Na}} = 11.01\%$. Model-brine row: $E_{\mathrm{Li}} = 51.63\%$ and $E_{\mathrm{Na}} = 9.97\%$.

PROCESS AND ECONOMICS

PrOMMiS/IDAES Uses Active Li And Na Transfer Terms

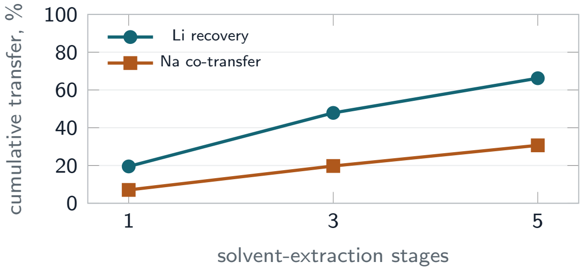
Smackover Li/Na feed → Divalent pretreatment → TBAC/DA DES + TOPO → ALAMO transfer surface → PrOMMiS/IDAES stages

Rezaee extraction responses are converted to Li/Na transfer variables for process calculations

Stages	Li recovery	Na co-transfer
1	19.508%	7.065%
3	47.850%	19.734%
5	66.212%	30.675%

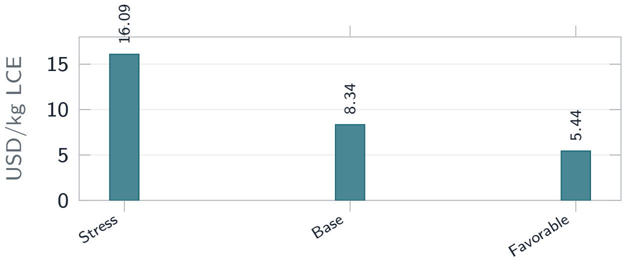
- The process layer uses PrOMMiS `SolventExtraction` objects with IDAES `MSContactor` units.
- Lithium and sodium material-transfer terms are active.
- Chloride transfer is held at zero inside the Li/Na extraction boundary.

Three Stages Recover 47.85% Lithium



- Smackover MS-2 at $O/A = 1$ and 10 wt% TOPO reaches 47.85% cumulative lithium recovery after three stages.
- Sodium co-transfer is 19.73% at the same condition.
- Five stages raise lithium recovery to 66.21% with a larger sodium burden.

Screening TEA Centers At 8.34 USD/kg LCE



- Base Smackover intensity is 8.34 USD/kg lithium carbonate equivalent.
- Stress case rises to 16.09 USD/kg through lower Li feed and lower O/A.
- Favorable case falls to 5.44 USD/kg through higher Li feed and higher O/A.

Economic Levers Are Feed Grade, Stages, And Losses

Scenario	Recovery	Cost	Product	Net
stress	25.04%	16.09	44.43	-0.18
base Smackover	47.85%	8.34	142.65	1.66
favorable	71.97%	5.44	383.13	11.33

Cost in USD/kg LCE; product in t/y; net in million USD/y.

Screening boundary

The TEA compares operating sensitivities for prioritization; it does not replace equipment design, residence-time sizing, solvent-loss experiments, or product-pricing analysis.

- Feed lithium grade is the strongest favorable lever.
- O/A ratio shifts recovery and solvent-contacting cost together.
- The stress case is not attractive at the assumed price basis.

THERMODYNAMIC EXTENSION

ePC-SAFT Supplies Activity And Fugacity Terms

$$\tilde{a}^{\mathrm{res}} = \tilde{a}^{\mathrm{hc}} + \tilde{a}^{\mathrm{disp}} + \tilde{a}^{\mathrm{assoc}} + \tilde{a}^{\mathrm{DH}} + \tilde{a}^{\mathrm{Born}}$$

$$\tilde{a}^{\mathrm{DH}} = -\frac{\kappa e^2}{12\pi\varepsilon_0\varepsilon_r k_B T}\sum_i x_i z_i^2 \chi_i$$

$$\kappa = \sqrt{\frac{\rho e^2}{k_B T \varepsilon_0 \varepsilon_r}\sum_j x_j z_j^2}$$

$$\chi_i = \frac{3}{(\kappa d_i)^3}\left[\frac{3}{2} + \ln(1+\kappa d_i) - 2(1+\kappa d_i) + \frac{1}{2}(1+\kappa d_i)^2\right]$$

$$\tilde{a}^{\mathrm{Born}} = -\frac{e^2}{4\pi\varepsilon_0 k_B T}\sum_i x_i z_i^2 \sum_{\delta\in\mathcal{D}}\left(1 - \frac{1}{\varepsilon_{r,m_\delta}}\right) D_{i,\delta}^{(m_\delta)}$$

$$D_{i,\mathrm{Born}}^{(\mathrm{bulk})} = \frac{1}{d_i^{\mathrm{Born}}}$$

$$D_{i,\mathrm{SSM}}^{(\mathrm{bulk})} = \frac{1}{d_i^{\mathrm{Born}} + \Delta d_i} - \frac{1}{d_i^{\mathrm{Born}}}$$

$$D_{i,\mathrm{DS}}^{(\mathrm{ion})} = \frac{1}{d_i^{\mathrm{Born}}} - \frac{1}{d_i^{\mathrm{Born}} + \Delta d_i}$$

residual
Helmholtz energy

↓

μ_i^{res}/RT and Z
from composition and
density derivatives

↓

$\ln\phi_i$ and $\ln\gamma_i$
fugacity and activity terms

↓

reaction quotient Q
and Li/Na transfer rows

$$\tilde{\mu}_k^{\mathrm{res}} = \tilde{a}^{\mathrm{res}} + Z - 1 + \left(\frac{\partial \tilde{a}^{\mathrm{res}}}{\partial x_k}\right)_{T,v,x_j}$$

$$\ln\varphi_k = \tilde{\mu}_k^{\mathrm{res}} - \ln Z, \qquad a_i = \gamma_i x_i, \qquad Q = \prod_i a_i^{\nu_i}$$

Shown terms are the Debye–Huckel finite-ion-size and modular Born inverse-diameter forms used for Li/Na activity and fugacity terms.

Case-Study Coverage Matrix

Lane	Current basis	Process-facing use
feed basis	Smackover MS-2 and high-Li Smackover sensitivity	defines the main and feed-grade sensitivity cases
paper-response bridge	Rezaee Li/Na extraction responses with O/A conversion	supplies D_{Li}, D_{Na}, and $S_{\mathrm{Li/Na}}$ for staged extraction
ALAMO-ready surrogate	current target-row fit for $\log D_{\mathrm{Li}}$ and $\log D_{\mathrm{Na}}$	provides the compact transfer surface used by the process model
PrOMMiS/IDAES stages	PrOMMiS/IDAES staged-extraction terms for Li and Na across 1, 3, and 5 stages	converts transfer variables to outlet recovery and raffinate behavior
screening TEA/UQ	stress, base, and favorable cases from the refreshed transfer surface	ranks operating levers for integration approval
direct ePC-SAFT transfer path	`b4144c72` phase-tagged cross-phase route; 26 Rezaee rows evaluated	supplies the activity/fugacity layer for replacing the paper-response bridge

Process-facing rows remain stable as the thermodynamic basis is exchanged.

The case study is a screening-quality bridge from Smackover feed chemistry to PrOMMiS/IDAES staged extraction and TEA, with ePC-SAFT activity and fugacity terms ready for direct process integration.

Technical Message

1. Smackover MS-2 is the main produced-water feed; the high-Li Smackover row defines the feed-grade sensitivity.
2. TBAC/DA DES + 10 wt% TOPO maps lithium and sodium transfer through ALAMO and into staged solvent extraction.
3. Three stages recover 47.85% lithium before the TEA pretreatment-loss adjustment.
4. The base screening TEA is 8.34 USD/kg LCE, with feed grade and phase ratio controlling the largest movement.
5. Ask: approve an internal ePC-SAFT-to-PrOMMiS/IDAES integration and validation sprint.

Supplemental Extension Context

Family	Why it belongs near produced water	Needed before modeling
CMM/REE ions	oilfield brines can carry critical-mineral value beyond Li	measured composition, aqueous speciation, and ion parameters
divalent ions	Ca, Mg, and Sr set pretreatment load and leakage risk	pretreatment data and extraction leakage bounds
hydrocarbons	oil-bearing water adds phase behavior and solvent-contact risks	oil-phase assay and phase-equilibrium data
ePC-SAFT extension	electrolyte and Born/dielectric terms give a thermodynamic route	accepted parameters and phase data for each species family

Boundary

Li/Na extraction remains the present case. CMM, REE, and hydrocarbon rows define where additional data would expand the same structure.

Use

The extension list guides what to measure before adding species, reactions, or oil-phase terms to the thermodynamic model.

USGS Energy Waste Science; critical-mineral produced-water review; electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature.

Sources

- Rezaee et al. 2025: TBAC/DA DES + TOPO extraction response basis.
- Rezaee et al. 2026: PC-SAFT/ePC-SAFT parameter and reaction-equilibrium support.
- U.S. Geological Survey southern Arkansas Smackover brine data release, DOI: `10.5066/P9QPRYZN`.
- USGS Energy Waste Science and produced-water lithium/mineral recovery context.
- Critical-mineral source potential review for U.S. oil and gas produced waters.
- Electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature for the extension basis.
- Produced-water feed table, surrogate fit table, PrOMMiS/IDAES staged-extraction table, and screening TEA table from the case-study result tables.